

Vectors: Triple Product

Vector Triple Product is a branch in vector algebra where we deal with the dot and cross multiplication of three vectors $\vec{A}, \vec{B}, \vec{C}$. The value of the vector triple product can be found by the cross product of a vector with the cross product of the other two vectors. It produces meaningful products of the form: $\vec{A} \cdot (\vec{B} \times \vec{C}), \vec{A} \cdot (\vec{B} \times \vec{C}), \vec{A} \times (\vec{B} \times \vec{C})$.

Scalar Triple product:

The scalar triple product, as its name may suggest, results in a scalar as its result. It is a means of combining three vectors via cross product and a dot product. Given the vectors

$$\mathbf{A} = A_1 \mathbf{i} + A_2 \mathbf{j} + A_3 \mathbf{k}$$

$$\mathbf{B} = B_1 \mathbf{i} + B_2 \mathbf{j} + B_3 \mathbf{k}$$

$$\mathbf{C} = C_1 \mathbf{i} + C_2 \mathbf{j} + C_3 \mathbf{k}$$

a scalar triple product will involve a dot product and a cross product

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$$

It is necessary to perform the cross product before the dot product when computing a scalar triple product,

$$\mathbf{B} \times \mathbf{C} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = \mathbf{i} \begin{vmatrix} B_2 & B_3 \\ C_2 & C_3 \end{vmatrix} - \mathbf{j} \begin{vmatrix} B_1 & B_3 \\ C_1 & C_3 \end{vmatrix} + \mathbf{k} \begin{vmatrix} B_1 & B_2 \\ C_1 & C_2 \end{vmatrix}$$

since $\mathbf{A} = A_1 \mathbf{i} + A_2 \mathbf{j} + A_3 \mathbf{k}$ one can take the dot product to find that

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (A_1) \begin{vmatrix} B_2 & B_3 \\ C_2 & C_3 \end{vmatrix} - (A_2) \begin{vmatrix} B_1 & B_3 \\ C_1 & C_3 \end{vmatrix} + (A_3) \begin{vmatrix} B_1 & B_2 \\ C_1 & C_2 \end{vmatrix}$$

which is simply

Important Formula 3.1.

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}$$

The product $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ is sometimes called the *scalar triple product* or *box product* and may be denoted by $[\mathbf{ABC}]$. The product $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ is called the *vector triple product*.

$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) =$ volume of a parallelepiped having $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ as edges, or the negative of this volume, according as $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ do or do not form a right-handed system. If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$ and $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$, then

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}$$

Example 3.1.2. Prove that

Important Formula 3.2.

$$\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = \mathbf{A} \times \mathbf{B} \cdot \mathbf{C}$$

Solution:

Notice that there are no brackets given here as the only way to evaluate the scalar triple products is to perform the cross products before performing the dot products^a. Let

$$\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$$

$$\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$$

$$\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$$

now,

$$\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = - \begin{vmatrix} C_1 & C_2 & C_3 \\ B_1 & B_2 & B_3 \\ A_1 & A_2 & A_3 \end{vmatrix} = \begin{vmatrix} C_1 & C_2 & C_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} = \mathbf{C} \cdot \mathbf{A} \times \mathbf{B} = \mathbf{A} \times \mathbf{B} \cdot \mathbf{C}$$

Vector Triple product:

The vector triple product, as its name suggests, produces a vector. It is the result of taking the cross product of one vector with the cross product of two other vectors.

Important Formula 3.3 (Vector Triple Product).

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$$

Proving the vector triple product formula can be done in a number of ways. The straightforward method is to assign

$$\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$$

$$\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$$

$$\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$$

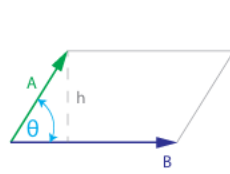
Formulas:

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$$

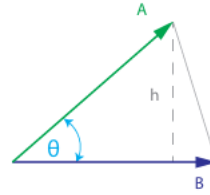
$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{B} \cdot \mathbf{C})\mathbf{A}$$

Area and Volume Using a Cross Product

The area of a parallelogram.



(a) The parallelogram



(b) The triangle

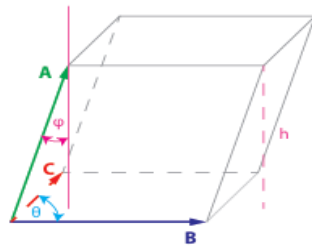
The area of a parallelogram is simply given by the product of the base and the height of the parallelogram. Here this is given by

$$\begin{aligned}
 \text{Area of parallelogram} &= h |\mathbf{B}| \\
 &= (|\mathbf{A}| \sin(\theta)) |\mathbf{B}| \\
 &= |\mathbf{A}| |\mathbf{B}| \sin(\theta) \\
 &= |\mathbf{A} \times \mathbf{B}|
 \end{aligned}$$

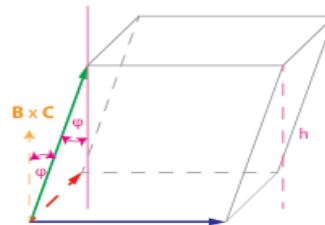
The area of a triangle is half the base times the height. From the figure we have

$$\begin{aligned}
 \text{Area of triangle} &= \frac{1}{2} |\mathbf{B}| h = \frac{1}{2} |\mathbf{B}| (|\mathbf{A}| \sin(\theta)) \\
 &= \frac{1}{2} |\mathbf{A} \times \mathbf{B}|
 \end{aligned}$$

The volume of a parallelepiped.



(a) The angle between $\mathbf{B}$ and $\mathbf{C}$ is θ .



(b) The angle between h and $\mathbf{A}$ is ϕ .

A parallelepiped is a three-dimensional object whose six sides are parallelograms. The volume of a parallelepiped is given by

$$V = (\text{Base Area})(\text{Height})$$

The area of the base is the area of a parallelogram as such one has

$$\text{Area of the Base} = |\mathbf{B} \times \mathbf{C}|$$

The height h requires a little geometry but is simply

$$h = |\mathbf{A}| \cos(\phi)$$

notice that the vector $\mathbf{B} \times \mathbf{A}$ is parallel to the line h . Thus the vector $\mathbf{A}$ makes an angle^a with the vector $\mathbf{B} \times \mathbf{C}$ of $\gamma = \phi$. Finally we have the volume of the parallelepiped given by

$$\begin{aligned} \text{Volume of parallelepiped} &= (\text{Base})(\text{height}) \\ &= (|\mathbf{B} \times \mathbf{C}|) (|\mathbf{A}| \cos(\gamma)) \\ &= |\mathbf{A}| |\mathbf{B} \times \mathbf{C}| \cos(\gamma) \\ &= |\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})| \end{aligned}$$

Solved Problems:

Question 1.

If $\mathbf{A} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{B} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$, and $\mathbf{C} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$, find (a) $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$, (b) $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$.

$$(a) \quad \mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix} = -\mathbf{i} + 7\mathbf{j} + 5\mathbf{k}.$$

$$\text{Then } (\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = (-\mathbf{i} + 7\mathbf{j} + 5\mathbf{k}) \times (\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 7 & 5 \\ 1 & -2 & 2 \end{vmatrix} = 24\mathbf{i} + 7\mathbf{j} - 5\mathbf{k}.$$

$$(b) \quad \mathbf{B} \times \mathbf{C} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -1 \\ 1 & -2 & 2 \end{vmatrix} = 0\mathbf{i} - 5\mathbf{j} - 5\mathbf{k} = -5\mathbf{j} - 5\mathbf{k}.$$

$$\text{Then } \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (3\mathbf{i} - \mathbf{j} + 2\mathbf{k}) \times (-5\mathbf{j} - 5\mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 2 \\ 0 & -5 & -5 \end{vmatrix} = 15\mathbf{i} + 15\mathbf{j} - 15\mathbf{k}.$$

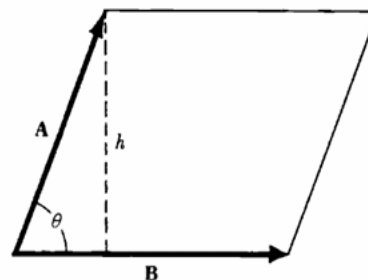
Thus $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} \neq \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$, showing the need for parentheses in $\mathbf{A} \times \mathbf{B} \times \mathbf{C}$ to avoid ambiguity.

Question 2.

Prove that the area of a parallelogram with sides $\mathbf{A}$ and $\mathbf{B}$ is $|\mathbf{A} \times \mathbf{B}|$.

$$\begin{aligned}\text{Area of parallelogram} &= h |\mathbf{B}| \\ &= |\mathbf{A}| \sin \theta |\mathbf{B}| \\ &= |\mathbf{A} \times \mathbf{B}|.\end{aligned}$$

Note that the area of the triangle with sides $\mathbf{A}$ and $\mathbf{B}$ is $\frac{1}{2} |\mathbf{A} \times \mathbf{B}|$.



Question 3.

Determine a unit vector perpendicular to the plane of $\mathbf{A} = 2\mathbf{i} - 6\mathbf{j} - 3\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$.

$\mathbf{A} \times \mathbf{B}$ is a vector perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$.

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -6 & -3 \\ 4 & 3 & -1 \end{vmatrix} = 15\mathbf{i} - 10\mathbf{j} + 30\mathbf{k}$$

$$\text{A unit vector parallel to } \mathbf{A} \times \mathbf{B} \text{ is } \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|} = \frac{15\mathbf{i} - 10\mathbf{j} + 30\mathbf{k}}{\sqrt{(15)^2 + (-10)^2 + (30)^2}} = \frac{3}{7}\mathbf{i} - \frac{2}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}.$$

Another unit vector, opposite in direction, is $(-3\mathbf{i} + 2\mathbf{j} - 6\mathbf{k})/7$.

Question 4.

If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$ show that

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}$$

$$\begin{aligned}\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) &= \mathbf{A} \cdot \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} \\ &= (A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}) \cdot [(B_2C_3 - B_3C_2)\mathbf{i} + (B_3C_1 - B_1C_3)\mathbf{j} + (B_1C_2 - B_2C_1)\mathbf{k}] \\ &= A_1(B_2C_3 - B_3C_2) + A_2(B_3C_1 - B_1C_3) + A_3(B_1C_2 - B_2C_1) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}\end{aligned}$$

Question 5.

Evaluate $(2\mathbf{i} - 3\mathbf{j}) \cdot [(\mathbf{i} + \mathbf{j} - \mathbf{k}) \times (3\mathbf{i} - \mathbf{k})]$.

$$\text{the result is } \begin{vmatrix} 2 & -3 & 0 \\ 1 & 1 & -1 \\ 3 & 0 & -1 \end{vmatrix} = 4.$$

Question 6.

Prove that $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$.

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}.$$

By a theorem of determinants which states that interchange of two rows of a determinant changes its sign, we have

$$\begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = - \begin{vmatrix} B_1 & B_2 & B_3 \\ A_1 & A_2 & A_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = \begin{vmatrix} B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \\ A_1 & A_2 & A_3 \end{vmatrix} = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A})$$

$$\begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = - \begin{vmatrix} C_1 & C_2 & C_3 \\ B_1 & B_2 & B_3 \\ A_1 & A_2 & A_3 \end{vmatrix} = \begin{vmatrix} C_1 & C_2 & C_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

Note:

If $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are coplanar the volume of the parallelepiped formed by them is zero.

Question 7.

Prove: (a) $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$, (b) $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C})$.

(a) Let $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$.

$$\begin{aligned} \text{Then } \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= (A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}) \times \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} \\ &= (A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}) \times ([B_2C_3 - B_3C_2]\mathbf{i} + [B_3C_1 - B_1C_3]\mathbf{j} + [B_1C_2 - B_2C_1]\mathbf{k}) \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_1 & A_2 & A_3 \\ B_2C_3 - B_3C_2 & B_3C_1 - B_1C_3 & B_1C_2 - B_2C_1 \end{vmatrix} \\ &= (A_2B_1C_2 - A_2B_2C_1 - A_3B_3C_1 + A_3B_1C_3)\mathbf{i} + (A_3B_2C_3 - A_3B_3C_2 - A_1B_1C_2 + A_1B_2C_1)\mathbf{j} \\ &\quad + (A_1B_3C_1 - A_1B_1C_3 - A_2B_2C_3 + A_2B_3C_2)\mathbf{k} \end{aligned}$$

Also $\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$

$$\begin{aligned} &= (B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k})(A_1C_1 + A_2C_2 + A_3C_3) - (C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k})(A_1B_1 + A_2B_2 + A_3B_3) \\ &= (A_2B_1C_2 + A_3B_1C_3 - A_2C_1B_2 - A_3C_1B_3)\mathbf{i} + (B_2A_1C_1 + B_2A_3C_3 - C_2A_1B_1 - C_2A_3B_3)\mathbf{j} \\ &\quad + (B_3A_1C_1 + B_3A_2C_2 - C_3A_1B_1 - C_3A_2B_2)\mathbf{k} \end{aligned}$$

and the result follows.

Question 8.

Prove: $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})$.

Question 9.

Prove: $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) + \mathbf{B} \times (\mathbf{C} \times \mathbf{A}) + \mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = \mathbf{0}$.

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

$$\mathbf{B} \times (\mathbf{C} \times \mathbf{A}) = \mathbf{C}(\mathbf{B} \cdot \mathbf{A}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C})$$

$$\mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\mathbf{C} \cdot \mathbf{B}) - \mathbf{B}(\mathbf{C} \cdot \mathbf{A})$$

Adding, the result follows.

Question 10.

Prove: $(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C} \times \mathbf{D}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C} \times \mathbf{D}) = \mathbf{C}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{D}) - \mathbf{D}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C})$.

$\mathbf{X} \times (\mathbf{C} \times \mathbf{D}) = \mathbf{C}(\mathbf{X} \cdot \mathbf{D}) - \mathbf{D}(\mathbf{X} \cdot \mathbf{C})$. Let $\mathbf{X} = \mathbf{A} \times \mathbf{B}$; then

$$(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D}) = \mathbf{C}(\mathbf{A} \times \mathbf{B} \cdot \mathbf{D}) - \mathbf{D}(\mathbf{A} \times \mathbf{B} \cdot \mathbf{C})$$

$$= \mathbf{C}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{D}) - \mathbf{D}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C})$$

$(\mathbf{A} \times \mathbf{B}) \times \mathbf{Y} = \mathbf{B}(\mathbf{A} \cdot \mathbf{Y}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{Y})$. Let $\mathbf{Y} = \mathbf{C} \times \mathbf{D}$; then

$$(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C} \times \mathbf{D}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C} \times \mathbf{D})$$

Question 11.

Prove: $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{B} \times \mathbf{C}) \times (\mathbf{C} \times \mathbf{A}) = (\mathbf{A} \cdot \mathbf{B} \times \mathbf{C})^2$.

Exercise:

1.

If $\mathbf{A} = \mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{B} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{C} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, find:

(a) $|(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}|$, (c) $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$, (e) $(\mathbf{A} \times \mathbf{B}) \times (\mathbf{B} \times \mathbf{C})$

(b) $|\mathbf{A} \times (\mathbf{B} \times \mathbf{C})|$, (d) $(\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$, (f) $(\mathbf{A} \times \mathbf{B})(\mathbf{B} \cdot \mathbf{C})$

Ans. (a) $5\sqrt{26}$, (b) $3\sqrt{10}$, (c) -20 , (d) -20 , (e) $-40\mathbf{i} - 20\mathbf{j} + 20\mathbf{k}$, ((f) $35\mathbf{i} - 35\mathbf{j} + 35\mathbf{k}$

2. Find the area of a parallelogram having diagonals $\bar{A} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\bar{B} = \hat{i} - 3\hat{j} + 4\hat{k}$.

Ans: $5\sqrt{3}$

3. Find the area of a triangle with vertices at $(3, -1, 2)$, $(1, -1, -3)$ and $(4, -3, 1)$.

Ans: $\frac{1}{2}\sqrt{165}$

4. Find the volume of the parallelepiped whose edges are represented by $\bar{A} = 2\hat{i} - 3\hat{j} + 4\hat{k}$, $\bar{B} = \hat{i} + 2\hat{j} - \hat{k}$, $\bar{C} = 3\hat{i} - \hat{j} + 2\hat{k}$. Ans: 7

5. Find the constant a such that the vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} - 3\hat{k}$, $3\hat{i} + a\hat{j} + 5\hat{k}$ are coplanar. Ans: $a = -4$